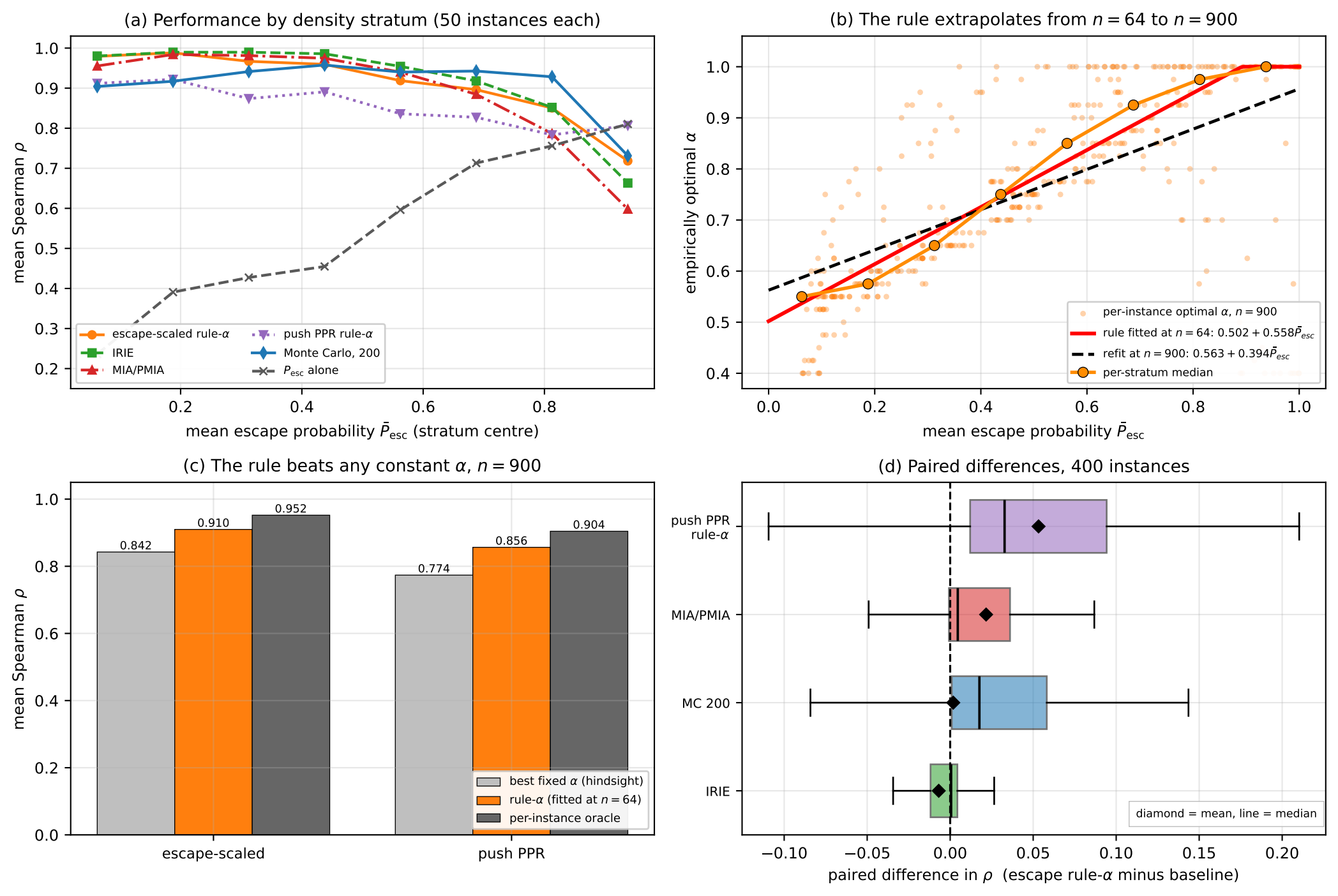
(a) Performance by density stratum (50 instances each)
mean Spearman ρ
mean escape probability $\bar{P}_{esc}$ (stratum centre)
escape-scaled rule-α
IRIE
MIA/PMIA
push PPR rule-α
Monte Carlo, 200
P_{esc} alone
(b) The rule extrapolates from $n = 64$ to $n = 900$
empirically optimal α
mean escape probability $\bar{P}_{esc}$
per-instance optimal α, $n = 900$
rule fitted at $n = 64$: $0.502 + 0.558\bar{P}_{esc}$
refit at $n = 900$: $0.563 + 0.394\bar{P}_{esc}$
per-stratum median
(c) The rule beats any constant α, $n = 900$
mean Spearman ρ
0.842
0.910
0.952
0.774
0.856
0.904
escape-scaled
push PPR
best fixed α (hindsight)
rule-α (fitted at $n = 64$)
per-instance oracle
(d) Paired differences, 400 instances
push PPR rule-α
MIA/PMIA
MC 200
IRIE
diamond = mean, line = median
paired difference in ρ (escape rule-α minus baseline)